## ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 2 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 3 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 4 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 5 | **(a)** 3 x 10^5 | 2 x 10^30 / 6 x 10^24. |
| 6 | **(a)** 50 per cent | 30/60. |
| 7 | **(d)** 2 : 1 | 75/25 = 3; 60/40 = 1.5; ratio 2 : 1. |
| 8 | **(a)** $S^{2-}$ | Z = 16 is sulphur, 2 extra electrons. |
| 9 | **(b)** the theory is strongly supported but remains open to tests | Science is provisional. |
| 10 | **(a)** only one component sublimes | Different behaviour on heating. |
| 11 | **(a)** 4.4 g | Mass of gas escaped. |
| 12 | **(c)** 24.3 u | (1896 + 250 + 286)/100 = 24.32. |
| 13 | **(d)** A is false but R is true. | A is false (theories can be refined); R is true. |
| 14 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 15 | **(c)** A is true but R is false. | A is true; R is false because the gas escapes. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | Area = 96 square cm (1); About 2 per cent (1). | 1+1 |
| 17 | Mass of solution = 240 g (1); 24/240 x 100 = 10 per cent (1). | 1+1 |
| 18 | 4 g H needs 32 g O (1); Water = 36 g; 8 g of oxygen left (1). | 1+1 |
| 19 | 63x + 65(100 - x) = 6350 (1); x = 75, so the heavier isotope is 25 per cent (1). | 1+1 |
| 20 | A straight line shows proportionality (1); Easy to read the slope and spot errors (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Mean = 42.6/3 (1); = 14.2 cm (1); Greatest deviation = 0.1 cm (1). | 1+1+1 |
| 22 | 320 g contains 120 g salt and 200 g water (1); At 20 degrees Celsius 200 g water holds 50 g (1); Crystals = 120 - 50 = 70 g (1). | 1+1+1 |
| 23 | S: 1/32 = 0.03125 (1); O: 1.6/16 = 0.1 (1); Ratio about 1 : 3: $SO_3$ (1). | 1+1+1 |
| 24 | 2, 8, 3 (1); $M^{3+}$ (1); Mass = 54 + 48 = 102 u (1). | 1+1+1 |
| | **OR:** Shows how reliable the result is (1); e.g. 12.3 +/- 0.1 cm (1); Allows comparison with other results (1). | 1+1+1 |
| 25 | A: 5/55 = 9.1 per cent (1); B: 8/108 = 7.4 per cent (1); A is more concentrated (1). | 1+1+1 |
| | **OR:** Mg2+ and O2- attract more strongly (1); More energy to break the lattice (1); Hence higher melting point (about 2800 vs 801 degrees Celsius) (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|

| | | |
|---|---|---|
| **26** | Apparatus: thread, bob, stand, stopwatch, scale (1); Independent: length; dependent: time period (1); Constants: mass, small angle (1); Time 20 oscillations and repeat (1); Plot T against L and conclude that T increases with L (1). | 1+1+1+1+1 |
| | **OR:** NaCl 36 g stays dissolved at 20 degrees Celsius (solubility 36) (1); KNO3 remaining dissolved = 32 g (1); KNO3 crystallised = 110 - 32 = 78 g (1); Crystals are nearly pure KNO3 (1); Fractional crystallisation (1). | 1+1+1+1+1 |
| **27** | Run 1: O = 1.6 g, Mg : O = 3 : 2 (1); Run 2: O = 4.0 g, Mg : O = 3 : 2 (1); Same ratio in both runs (1); Formula MgO: 24 : 16 = 3 : 2 (1); Law and formula verified (1). | 1+1+1+1+1 |
| | **OR:** 69x + 71(100 - x) = 6980 (1); x = 60, so 60 per cent and 40 per cent (1); Neutrons: 38 and 40 (1); Element is gallium (1); Average mass between 69 and 71 (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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# THE MASTERMIND COACHING CLASSES

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## ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 2 | **(a)** 5.4 cm | Subtract the zero error. |
| 3 | **(b)** 32 g | Water 100 g holds 32 g; crystals 64 - 32. |
| 4 | **(b)** 3 : 8, $CO_2$ | Atoms ratio 2.4/12 : 6.4/16 = 1 : 2. |
| 5 | **(b)** 20 and 16 | X lost 2, Y gained 2. |
| 6 | **(a)** dosage of a medicine for a child | Safety critical. |
| 7 | **(a)** iron and sulphur | Only iron is magnetic. |
| 8 | **(b)** 78.5 | 52.5/47.5 x 71 = 78.5. |
| 9 | **(a)** 20 | 10x + 11(100 - x) = 1080. |
| 10 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 11 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 12 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 13 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 14 | **(d)** 1 per cent | 0.2/20.4. |
| 15 | **(b)** 18 g | 50 - 32. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | Boiling point is fixed (1); Heat supplies latent heat (1). | 1+1 |
| 17 | Na+: 2, 8; F-: 2, 8 (1); Same as neon (1). | 1+1 |
| 18 | Maximum 2n squared with n = 1 is 2 (1); K shell is the smallest and nearest to the nucleus (1). | 1+1 |
| 19 | Distance = 3 x 10^5 x 3.15 x 10^7 (1); = 9.45 x 10^12 km (1). | 1+1 |
| 20 | Slow cooling allows ordered growth (1); Few nuclei form (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | p + n = 31 and n = p + 5 (1); 2p = 26, so p = 13 and n = 18 (1); Z = 13; configuration 2, 8, 3 (1). | 1+1+1 |
| 22 | Rising curve (1); Not a straight line (1); Longer pendulum has longer period (1). | 1+1+1 |
| 23 | Salt gives ions that neutralise the charge on particles (1); Particles aggregate (1); Soap separates (1). | 1+1+1 |
| 24 | Molecular mass = 12 + 4 = 16 u (1); Carbon = 12/16 x 100 = 75 per cent (1); Hydrogen = 4/16 x 100 = 25 per cent (1). | 1+1+1 |
| | **OR:** Electrons occupy fixed shells (1); No radiation in these states (1); Energy changes only by jumps (1). | 1+1+1 |
| 25 | Mean = 54.4 s for 30 (1); T = 1.81 s (1); Spread about 0.4 / 54.4 = 0.7 per cent (1). | 1+1+1 |
| | **OR:** X = 0.27 (1); Y = 0.6, Z = 0.8 (1); X is most strongly adsorbed (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | Same mass and volume (1); Vary only temperature (1); Record times (1); Graph of time vs temperature (1); Errors: reaction time and stirring (1). | 1+1+1+1+1 |

| | | |
|---|---|---|
| | **OR:** Particles carry charge and repel (1); Coagulation with electrolytes (1); Electrophoresis moves particles in a field (1); Dialysis separates dissolved ions (1); Examples: blood purification (1). | 1+1+1+1+1 |
| **27** | Atoms of different elements have different masses (1); 12 g of Mg has half as many atoms as 12 g of C (1); Number of atoms is proportional to mass divided by atomic mass (1); 24 g of Mg has as many atoms as 12 g of C (1); Atomic masses link mass to number of atoms (1). | 1+1+1+1+1 |
| | **OR:** n = 24 - 12 = 12 (1); B: p = 11, n = 12, A = 23 (1); B is sodium $^{23}_{11}\mathrm{Na}$ (1); 2, 8, 1 (1); A is magnesium-24 (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 28 g | 28/44 of the mass is nitrogen. |
| 2 | **(d)** argon | 18 electrons. |
| 3 | **(a)** check the experiment and the assumptions in the model | Investigate. |
| 4 | **(b)** dissolve in water, filter, evaporate | Sugar dissolves. |
| 5 | **(c)** 27.3 per cent | 12/44 x 100 = 27.3. |
| 6 | **(c)** electron | An electron is about 1/1836 of the mass of a proton. |
| 7 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 8 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 9 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 10 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 11 | **(c)** 15 per cent | 3 x 5 per cent. |
| 12 | **(b)** the one with R_f 0.9 | Largest R_f. |
| 13 | **(c)** 310 u | 120 + 62 + 128. |
| 14 | **(a)** 5 | Z = 15: 2, 8, 5. |
| 15 | **(c)** to test the procedure and find problems | Trial run. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | 117/58.5 = 2 (1); Twice as many (1). | 1+1 |
| 17 | No dense positive centre in his model (1); It predicted only small deflections (1). | 1+1 |
| 18 | Mean = 9.76 m/s^2 (1); Range = 0.3 m/s^2 (1). | 1+1 |
| 19 | Solute may have low solubility (1); e.g. a saturated solution of chalk or CO2 in water is dilute (1). | 1+1 |
| 20 | Formula unit mass of CuO = 64 + 16 = 80 u (1); So 64 g of copper is present in 80 g (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Mass of O2 = 24.5 - 14.9 = 9.6 g (1); Percentage = 9.6/24.5 x 100 (1); = 39.2 per cent (1). | 1+1+1 |
| 22 | Same valence electrons in a group (1); Across a period valence electrons increase 1 to 8 (1); Chemical behaviour depends on valence electrons (1). | 1+1+1 |
| 23 | Volume = 144 cubic m (1); Mass = 144 x 1.2 (1); = about 173 kg (1). | 1+1+1 |
| 24 | Ink spots from document and suspects (1); Run chromatography (1); Compare bands and R_f values (1). | 1+1+1 |
| | **OR:** Weighing error (1); Leakage of a gas (1); The law holds when all products are weighed (1). | 1+1+1 |
| 25 | Splits on neutron capture giving energy (1); Chain reaction controlled in reactors (1); Radioactive waste and accidents (1). | 1+1+1 |
| | **OR:** Informed consent (1); Safety tests first (1); Honest reporting of side effects (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|

| | | |
|---|---|---|
| **26** | Strength: evidence-based (1); Strength: self-correcting (1); Limitation: cannot test untestable ideas (1); Chance: penicillin discovery (1); Persistence: many experiments repeated (1). | 1+1+1+1+1 |
| | **OR:** Spot unknown and known dyes on the same paper (1); Run in a suitable solvent (1); Compare R_f values (1); Same R_f means same dye (1); Solvent changes R_f (1). | 1+1+1+1+1 |
| **27** | Conservation: atoms rearranged (1); Constant proportions: fixed ratios of atoms (1); Atoms are divisible (1); Isotopes differ in mass (1); Atoms combine in whole-number ratios remains valid (1). | 1+1+1+1+1 |
| | **OR:** Closest approach of alpha particles gives an upper limit for the nucleus size (1); Atom radius about 10^-10 m (1); Density = mass/volume is huge (1); About 10^18 kg per cubic metre (1); Nucleus holds nearly all the mass (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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